
COMPTE RENDU : TP 1 – RECONNAISSANCE ET CARTOGRAPHIE RÉSEAU (NMAP)

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Environnement: Virtual Box (Kali Linux, Ubuntu Server, Windows)

Objectif : Maîtriser l'outil Nmap pour identifier les machines actives, les ports ouverts et les services d'un réseau cible.

1. Présentation de l'outil Nmap

Nmap (Network Mapper) est un outil de sécurité open-source utilisé pour l'exploration réseau et l'audit de sécurité. Dans une phase de test d'intrusion, il intervient lors de la **reconnaissance**. Son rôle est de déterminer :

- Quels hôtes sont "en vie" sur le réseau.
- Quels services (noms et versions) ces hôtes proposent.
- Quels systèmes d'exploitation ils utilisent.

2. Méthodologie et Analyse des Commandes

Voici le récapitulatif des commandes utilisées durant le TP et leurs fonctions :

Commande	Option	Utilité technique
<code>nmap -sn 192.168.56.0/24</code>	Ping Scan	Découvre les machines actives sans scanner les ports (rapide et discret).
<code>sudo nmap -sS [IP]</code>	SYN Scan	Scan "semi-ouvert" qui identifie les ports sans établir de connexion complète (discrétion).
<code>nmap -sV [IP]</code>	Service Version	Interroge les ports ouverts pour obtenir le nom et la version exacte des logiciels.
<code>sudo nmap -O [IP]</code>	OS Detection	Analyse la pile TCP/IP pour deviner le système d'exploitation installé.

<code>nmap -A [IP]</code>	Aggressive	Combine la détection d'OS, de version, le script scanning et le traceroute.
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3. Résultats Obtenus (Synthèse)

Après configuration du laboratoire (installation d'Apache et SSH sur Ubuntu), les résultats sont les suivants :

- **Cible 1 (Ubuntu - 192.168.56.10) :**
 - **Ports ouverts :** 22 (SSH), 80 (HTTP).
 - **Versions :** OpenSSH 8.2p1, Apache httpd 2.4.41.
 - **OS :** Linux (Noyau 4.x ou 5.x).
- **Cible 2 (Windows - 192.168.56.20) :**
 - **Statut :** Nécessite l'option `-Pn` si le pare-feu est actif ou une désactivation du firewall pour répondre au ping.

```

aliawy@kali: ~
File Actions Edit View Help
— 192.168.56.20 ping statistics —
8 packets transmitted, 0 received, +6 errors, 100% packet loss, time 7142ms
pipe 4

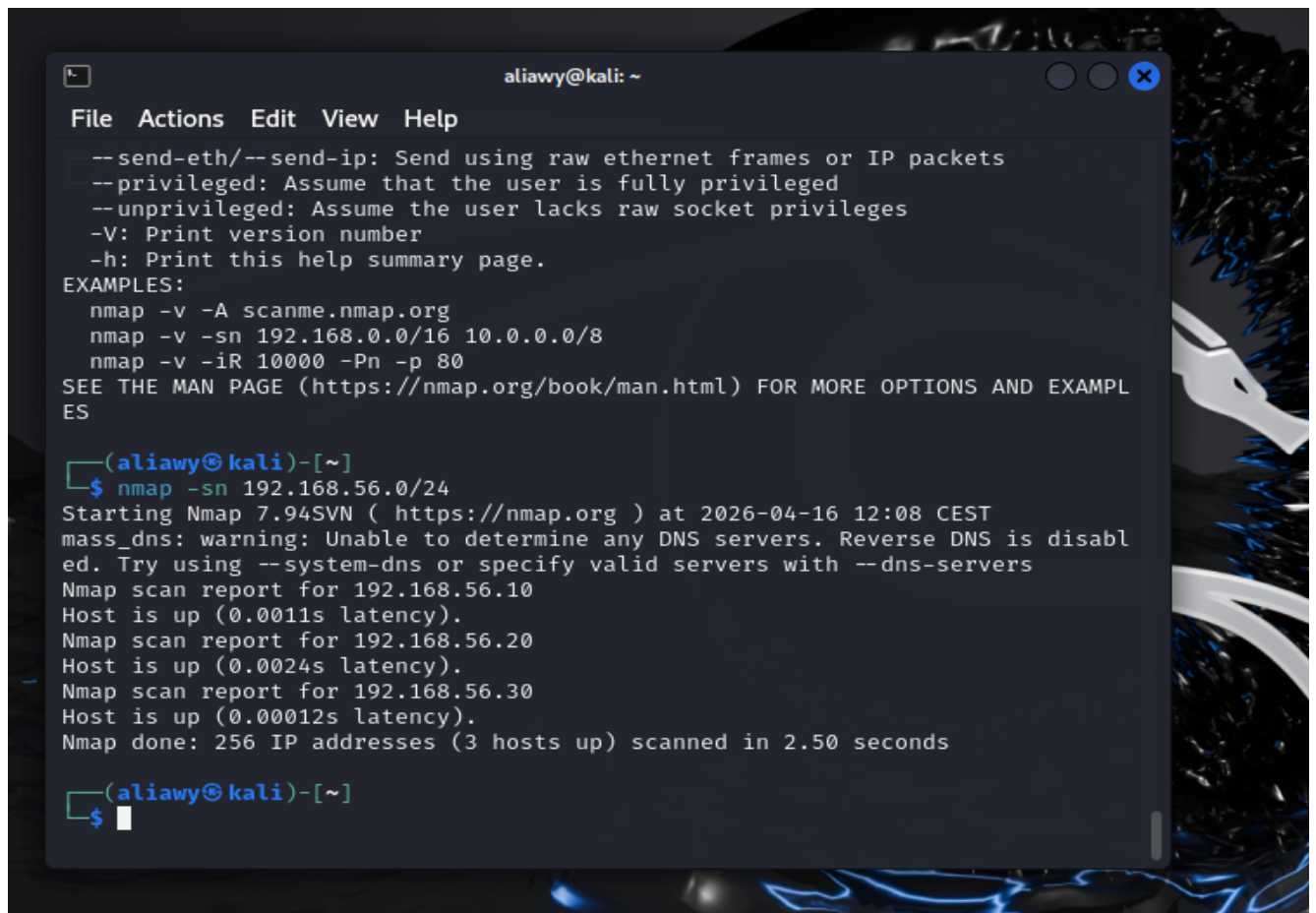
(aliawy@kali)-[~]
$ ping 192.168.56.20
PING 192.168.56.20 (192.168.56.20) 56(84) bytes of data.
64 bytes from 192.168.56.20: icmp_seq=1 ttl=128 time=2.06 ms
64 bytes from 192.168.56.20: icmp_seq=2 ttl=128 time=1.30 ms
^C
— 192.168.56.20 ping statistics —
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.300/1.678/2.056/0.378 ms

(aliawy@kali)-[~]
$ ping 192.168.56.10
PING 192.168.56.10 (192.168.56.10) 56(84) bytes of data.
64 bytes from 192.168.56.10: icmp_seq=1 ttl=64 time=1.07 ms
64 bytes from 192.168.56.10: icmp_seq=2 ttl=64 time=1.40 ms
64 bytes from 192.168.56.10: icmp_seq=3 ttl=64 time=0.940 ms
^C
— 192.168.56.10 ping statistics —
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 0.940/1.135/1.398/0.192 ms

(aliawy@kali)-[~]
$

```

Capture 1 : test de communication entre kali et ubuntu,windows

A terminal window titled 'aliawy@kali: ~' with a menu bar (File, Actions, Edit, View, Help). It displays the nmap help text, including options like --send-eth, --privileged, --unprivileged, -V, and -h. It also shows examples of nmap commands. The user then runs 'nmap -sn 192.168.56.0/24', which outputs a scan report for three hosts (192.168.56.10, 192.168.56.20, 192.168.56.30) and a summary: 'Nmap done: 256 IP addresses (3 hosts up) scanned in 2.50 seconds'.

```
aliawy@kali: ~
File Actions Edit View Help

--send-eth/--send-ip: Send using raw ethernet frames or IP packets
--privileged: Assume that the user is fully privileged
--unprivileged: Assume the user lacks raw socket privileges
-V: Print version number
-h: Print this help summary page.
EXAMPLES:
nmap -v -A scanme.nmap.org
nmap -v -sn 192.168.0.0/16 10.0.0.0/8
nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES

(aliawy@kali)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 12:08 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.0011s latency).
Nmap scan report for 192.168.56.20
Host is up (0.0024s latency).
Nmap scan report for 192.168.56.30
Host is up (0.00012s latency).
Nmap done: 256 IP addresses (3 hosts up) scanned in 2.50 seconds

(aliawy@kali)-[~]
$
```

Partie 1 : découverte des machines actives

```
aliawy@kali: ~  
File Actions Edit View Help  
ed. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.10  
Host is up (0.00096s latency).  
All 1000 scanned ports on 192.168.56.10 are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
MAC Address: 08:00:27:0F:2F:59 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.75 seconds  
  
(aliawy@kali)-[~]  
$ sudo nmap -sS 192.168.56.20  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 12:20 CEST  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl  
ed. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.20  
Host is up (0.0045s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 08:00:27:8B:BE:94 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.63 seconds  
  
(aliawy@kali)-[~]  
$  
  
aliawy@kali: ~  
File Actions Edit View Help  
Host is up (0.0023s latency).  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
MAC Address: 08:00:27:8B:BE:94 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 1.67 seconds  
  
(aliawy@kali)-[~]  
$ sudo nmap -sS 192.168.56.10  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 13:19 CEST  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabl  
ed. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.10  
Host is up (0.00069s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE  
22/tcp    open  ssh  
80/tcp    open  http  
MAC Address: 08:00:27:0F:2F:59 (Oracle VirtualBox virtual NIC)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds  
  
(aliawy@kali)-[~]  
$
```

Etape 2 : scan des ports

Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds

```
(aliawy@kali)-[~]
$ nmap -sv 192.168.56.10
Nmap 7.94SVN ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sl: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PY/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
```

```
File Actions Edit View Help
--traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
  Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports sequentially - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
  -sC: equivalent to --script-default
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of
```

```
--iflist: Print host interfaces and routes (for debugging)
--append-output: Append to rather than clobber specified output files
--resume <filename>: Resume an aborted scan
--noninteractive: Disable runtime interactions via keyboard
--stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
--webxml: Reference stylesheet from Nmap.Org for more portable XML
--no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
  -6: Enable IPv6 scanning
  -A: Enable OS detection, version detection, script scanning, and traceroute
  --datadir <dirname>: Specify custom Nmap data file location
  --send-eth/--send-ip: Send using raw ethernet frames or IP packets
  --privileged: Assume that the user is fully privileged
  --unprivileged: Assume the user lacks raw socket privileges
  -V: Print version number
  -h: Print this help summary page.
EXAMPLES:
  nmap -v -A scanme.nmap.org
  nmap -v -sn 192.168.0.0/16 10.0.0.0/8
  nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
Scantype v not supported
```

```
(aliawy@kali)-[~]
$
```

Partie3 : identification des services

```

(aliawy@kali)-[~]
$ sudo nmap -O 192.168.56.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 13:28 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.00083s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
MAC Address: 08:00:27:0F:2F:59 (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.8
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.39 seconds

(aliawy@kali)-[~]
$

```

Partie 4 : détection du système d'exploitation

```

(aliawy@kali)-[~]
$ nmap -A 192.168.56.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 13:31 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.0028s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.14 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 bf:f8:22:b6:6a:4c:0a:93:9f:a2:17:03:71:f5:c8:87 (ECDSA)
|_  256 a5:a1:c2:13:7e:2c:70:8f:8f:20:3c:74:3a:2c:87:a9 (ED25519)
80/tcp    open  http     Apache httpd 2.4.52 ((Ubuntu))
|_ http-server-header: Apache/2.4.52 (Ubuntu)
|_ http-title: Apache2 Ubuntu Default Page: It works
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.91 seconds

(aliawy@kali)-[~]
$

```

Partie 5 : scan avance

4. Analyse de Sécurité et Recommandations

L'utilisation de Nmap permet de mettre en évidence des failles potentielles :

1. **Divulgarion d'informations** : L'option `-sV` révèle les versions des logiciels. Si une version est obsolète (ex: Apache ancien), un attaquant peut chercher un "exploit" correspondant.
2. **Ports inutiles** : Tout port ouvert non indispensable (ex: un port de base de données exposé) doit être fermé pour réduire la surface d'attaque.
3. **Audit** : Un administrateur doit utiliser Nmap régulièrement pour s'assurer qu'aucun service non autorisé n'a été installé par un utilisateur.

5. Options Avancées (Piste d'approfondissement)

Pour aller plus loin, nous avons identifié trois options majeures :

- `-p-` : Pour scanner l'intégralité des 65 535 ports (au lieu des 1000 par défaut).

```
(aliawy@kali)-[~]
$ nmap -p- 192.168.56.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 14:43 CEST
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.0047s latency).
Not shown: 65533 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 20.03 seconds

(aliawy@kali)-[~]
$
```

- `--script vuln` : Pour lancer automatiquement une recherche de vulnérabilités connues

```
(aliawy@kali)-[~]
$ nmap --script vuln 192.168.56.10
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 14:46 CEST
Failed to resolve "script".
Failed to resolve "vuln".
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers
Nmap scan report for 192.168.56.10
Host is up (0.0045s latency).
Not shown: 998 closed tcp ports (conn-refused)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 0.65 seconds

(aliawy@kali)-[~]
$
```

- sur les services détectés.

- **-T4** : Pour accélérer la vitesse d'exécution du scan sur un réseau stable.

```
(aliawy@kali)-[~]  
$ nmap -T4 192.168.56.20  
Starting Nmap 7.94SVN ( https://nmap.org ) at 2026-04-16 14:50 CEST  
mass_dns: warning: Unable to determine any DNS servers. Reverse DNS is disabled. Try using --system-dns or specify valid servers with --dns-servers  
Nmap scan report for 192.168.56.20  
Host is up (0.0013s latency).  
Not shown: 997 closed tcp ports (conn-refused)  
PORT      STATE SERVICE  
135/tcp   open  msrpc  
139/tcp   open  netbios-ssn  
445/tcp   open  microsoft-ds  
  
Nmap done: 1 IP address (1 host up) scanned in 1.44 seconds
```

Conclusion

Ce TP a permis de comprendre l'importance de la phase de reconnaissance. La maîtrise de Nmap est indispensable pour tout expert en cyber sécurité afin de cartographier un périmètre et d'anticiper les vecteurs d'attaque.